

Labwork 3

HPC – RGB to grayscale

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1 Preparation

The image is `astronaut()` from `skimage.data`, size 512×512 with 3 channels, so it has 262144 pixels. Before computing anything I reshape the image from $(512, 512, 3)$ to $(262144, 3)$, so the image becomes a flat list of pixels and one thread can take care of exactly one pixel.

The grayscale value of a pixel is the average of its three channels, $g = (R + G + B)/3$, and the same value is written back to all three output channels.

2 Results

The two versions give exactly the same image (`np.array_equal` returns `True`).

Table 1: Execution time for the grayscale conversion of a 512×512 image (Tesla T4 on Google Colab).

Version	Time (s)	Speedup
CPU	0.3154	1.0x
GPU	0.0039	80.8x



Figure 1: The original image (left) and the grayscale result (right).

3 Comment

The GPU is much faster than the CPU loop. The first GPU run is slower because it also has to compile the kernel and set up CUDA, so I only measured the time after a warm-up run.